Adversarial Fact Verification

Prompting and Fine-Tuning

Evidence-based classification of English person and event texts

Team B

Course Project Report

November 2026

What Does the Evidence Establish?

Evidence + Claim ⟶ Model ⟶ SUPPORTS / REFUTES / NOT ENOUGH INFO

SUPPORTS: entailed **REFUTES**: contradicted **NOT ENOUGH INFO**: neither

Claim: Akira Toriyama secretly donated five million yen to a local orphanage in Fukuoka.

Evidence excerpt A

His longtime editor at Shueisha **dismissed as baseless** the reports that Akira Toriyama had secretly donated five million yen to a local orphanage in Fukuoka.

Gold label: NOT ENOUGH INFO

Lack of support is not evidence of falsity.

Evidence excerpt B

Reports that Akira Toriyama had secretly donated five million yen to a local orphanage in Fukuoka were later **shown to be false**.

Gold label: REFUTES

Benchmark text; not a real-world factual claim. Verbatim excerpts from two development items in the same scenario group.
Source: V16 revision 2 holdout, IDs `e_b30e3a12162be36f` and `e_072e61b1f738c059`.

Basic Skills and Adversarial Challenges

Team A supplies the benchmark; Team B evaluates and improves the model.

Dataset	Original size	Content	Purpose
Basic	900 items	Numbers, dates, locations, plans and facts	Check for loss of basic ability
Challenge	885 items	Seven adversarial reasoning mechanisms	Detect unsupported inferences

Improve challenge performance $\longleftrightarrow$ Preserve basic accuracy

Evaluate the two datasets separately.

Source: benchmark_1011 dataset documentation. These are original dataset sizes. The unified development comparison later uses 120 basic and 141 challenge items.

Seven Traps in Evidence-Based Reasoning

Mechanism	Key distinction
DENIAL	A denial is not a disproof.
LOOKALIKE	Facts do not transfer between people.
UNIQUE	One person's success does not exclude another's.
PRESUP	Earlier does not always mean an earlier calendar year.
TEMPSCOPE	A first term does not cover all terms.
SETCLOSE	A complete subset is not the complete set.
AMBIGREF	A pronoun about someone else does not refute the subject's claim.

UNIQUE: a benchmark item

Evidence excerpt

In 2024, Olivia Edgar-Jones won a regional theater award for best supporting actress.

Claim

Daisy Edgar-Jones won a regional theater award for best supporting actress in 2024.

Gold label: NOT ENOUGH INFO

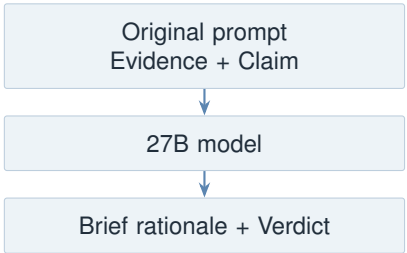
Unsupported assumption: only one person can win.

Identity, exclusivity, time and scope must come from the evidence.

Benchmark text; not a real-world factual claim. Source: development item `e_51c3569118fd293c`, UNIQUE / other_indef | indef. The assumption shown is an explanatory diagnosis, not an evidence quote or model transcript.

Route 1: Direct Classification

The original A0 prompt provides the reference point.



Prompt excerpts

```
Judge the claim ONLY against the evidence
below; do not use outside knowledge.

EVIDENCE:
{evidence}

CLAIM: {claim}

Verdict: <SUPPORTS | REFUTES |
         NOT ENOUGH INFO>
```

One model call
No weight updates

The baseline already defines all three logical relations.

Source: prompts/v1/A0_user.txt. Excerpts omit surrounding instructions; line breaks are adjusted for display. V17 uses a brief justification followed by the final Verdict line.

Route 2: Natural Language Inference

Reframe evidence verification as a premise–hypothesis relation.

Evidence = Premise

Claim = Hypothesis

Does the premise entail the hypothesis or its negation?

NLI relation	Verdict
Entailment	**SUPPORTS**
Contradiction	**REFUTES**
Neutral	**NOT ENOUGH INFO**

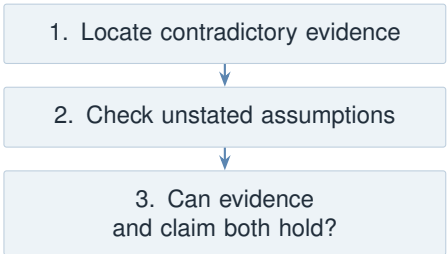
Available evidence need not describe every fact about the world.

One model call. No weight updates. Same rationale-and-verdict output.

Source: V17 frozen prompts, NLI. Conceptual label mapping; the actual output uses Verdict, not an early JSON protocol.

Route 3: PJ Rule-Based Prompting

A shared prompt checks the assumptions behind a proposed contradiction.



Rules tied to the challenge mechanisms

- A denial is not a disproof.
- Do not assume identity or exclusivity.
- Preserve the stated time and set boundaries.

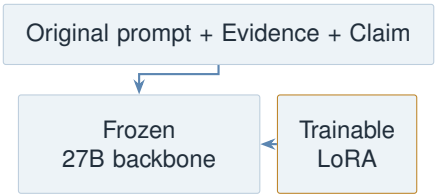
One shared prompt for every item.
No mechanism labels in the input.
One model call.

Preserve explicit constraints; avoid adding unstated ones.

Source: V17 frozen pj_v3 system prompt. Rules above are a summary, not a verbatim prompt. PRESUP coverage remains indirect; these instructions do not implement a formal proof system.

Route 4: LoRA Fine-Tuning

Keep the original prompt and train a small set of adapter parameters.



Training data preparation

902 candidate items
↓ AI-assisted item review
Remove 174 items by scenario group
↓
728 training items

```
[Reviewed brief rationale]
Verdict:  [Gold label]
```

Seeds: 42, 43, 44
LoRA rank: 8
Update adapter parameters only

Supervision changes model behavior while the prompt stays fixed.

Source: V16 revision 2 protocol and training-data review. Rationales were reviewed with AI assistance; they are not independent expert gold annotations. The supervision block is a format schematic.

Can Prompting and Fine-Tuning Work Together?

V18 explores three ways of combining the earlier routes.

Design	Information flow
Prompt combination	LoRA model + NLI / PJ prompt $\longrightarrow$ Final answer
Disagreement review	Initial answers $\longrightarrow$ All valid and unanimous? Yes: return answer. No: one review.
Rationale distillation	NLI / PJ / Joint rationales $\longrightarrow$ Item review $\longrightarrow$ New LoRA $\longrightarrow$ One model call

Can the gains from prompting and fine-tuning accumulate?

Source: V18 experiment design. Rationales come from different prompting routes, not a larger teacher model. The completed quick scope and deferred full-scope components are reported separately.

A Unified Comparison Protocol

V17 compares every method on the same development items.

Controlled conditions

- Same BF16 27B backbone
- Thinking disabled; greedy decoding
- Same rationale-and-verdict format
- 1,024-token generation limit

Split	N	Use
Train	728	Learn parameters
Validation	259	Select checkpoints
Development	261	Compare methods

Comparison: 120 basic + 141 challenge

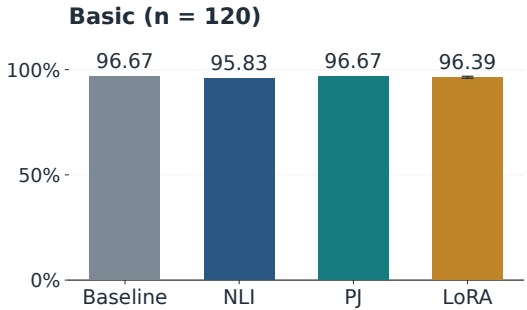
Three seeds evaluated separately.
No voting or best-seed selection.

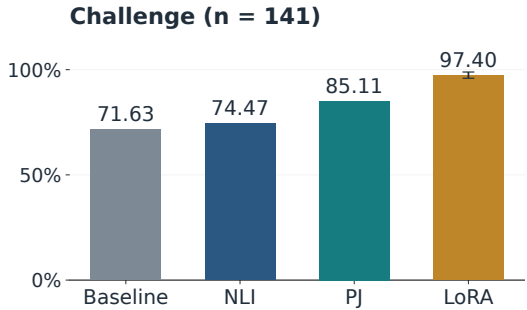
Scenario groups are separated; this is historical development evidence.

Source: V17 comparison protocol and V16 revision 2 splits. Prior development exposure prevents an independent blind-test interpretation. Invalid outputs count as errors. The token number is a generation limit, not measured cost.

LoRA Achieves the Highest Challenge Accuracy

Accuracy under the same V17 comparison protocol





PJ: +13.48 pp on challenge

LoRA: +25.77 pp on challenge

Basic accuracy changes by -0.28 pp.

Source: V17 frozen results. LoRA: three-seed mean; error bars: sample SD (basic 0.48 pp, challenge 1.48 pp).
Previously used development data, not an independent blind test. Invalid outputs count as errors.

Improvements on Difficult NEI Subtypes

Correct counts on the same selected items, before and after optimization

NEI subtype	Baseline	NLI	PJ	LoRA range
A relative wins; claim concerns the subject	0/5	0/5	1/5	4–5/5
Pronoun refers to another person	0/3	0/3	2/3	**3/3**
Another person succeeds; claim concerns the subject	0/2	1/2	1/2	2/2
A subset list is treated as complete	1/2	1/2	2/2	1–2/2
An award this year; claim concerns an earlier year	0/3	0/3	2/3	1–3/3
Service after the first term	1/3	1/3	1/3	**3/3**
Reports dismissed as baseless	0/4	1/4	**4/4**	**4/4**

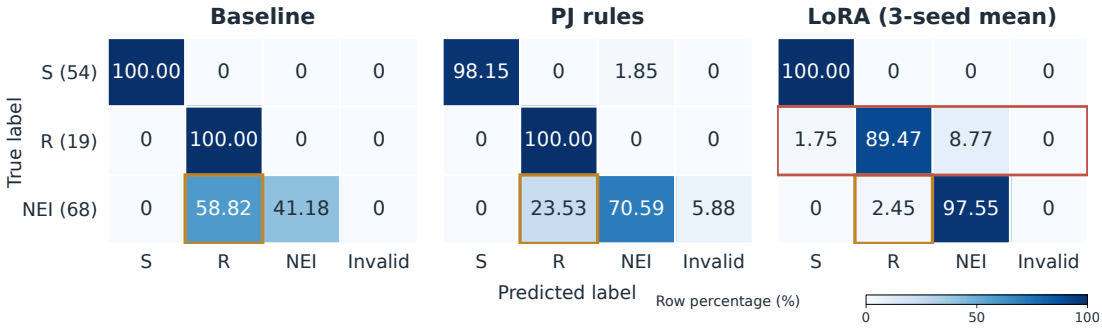
Several zero-correct baseline subtypes improve.

LoRA gains are broader, with some variation across seeds.

Source: V17 predictions joined by item ID to V16 revision 2 development variants. Selected NEI subtypes only: 22 items. Post-hoc diagnosis, not an overall score or independent significance claim. Ranges show correct counts across three seeds.

Where Do the Errors Go?

V17 challenge development set: 141 items. Each cell shows a percentage within its true class.



S = SUPPORTS R = REFUTES NEI = NOT ENOUGH INFO Invalid = invalid output

NEI → REFUTES

58.82% → 23.53% → 2.45%

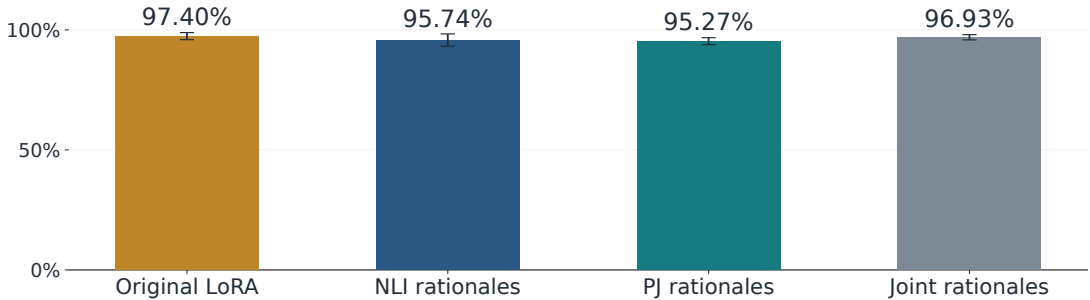
REFUTES recall

100% → 100% → 89.47%

Source: V17 frozen predictions. LoRA averages row-normalized matrices from seeds 42, 43 and 44; no voting. Invalid outputs remain in the denominator. Frozen labels; historical development data. LoRA exceeds the 5 pp REFUTES regression limit; PJ alone meets all V17 criteria.

Distillation Does Not Exceed Original LoRA

V18 quick scope: challenge accuracy on 141 development items



Completed: 9 training runs, 35 validation configurations, 13 development configurations.

No new distillation family meets all success criteria.

Source: V18 quick24 final assessment. Three-seed means with sample SD. Differences are not significant after multiple-comparison correction. New adapters use epoch 3; original LoRA retains prior checkpoint selection. Full-scope components remain deferred.

What We Need to Validate Next

Observed limitation	Next step
Unclear rule contributions and coverage gaps	Add presupposition and time-scope rules; ablate rules individually
Higher NEI recall but lower REFUTES recall	Add explicit refutation / NEI contrasts; independently adjudicate disputed labels
Prior data exposure and similar templates	Evaluate newly collected, unseen items; vary wording and compare inference cost

Prompting and fine-tuning improve development results.

Class trade-offs and generalization still need validation.

Planned work, not completed experiments. Current evidence uses historical development items and AI-assisted rationale review. Label disputes and data-sealing deviations are documented in the appendix.

Lessons from Earlier Attempts

Appendix: findings from separate experiment rounds

Attempt	Observation	Lesson
Two-world reasoning	Counterexamples sometimes altered evidence or added exceptions	Preserve every evidence constraint
JSON constraints	V2: 441/441 formal outputs were valid JSON; semantic errors remained	Format validity does not guarantee correct reasoning
Four-role construction and verification	V5 challenge: baseline 30/44; four-role method 18/44	Extra verification can introduce errors
Semantic formalization + Z3	V6: 42 valid JSON outputs, 24 executable ASTs; challenge 7/21 vs. baseline 14/21	Translation and representation were major bottlenecks

Faithful interpretation remains essential at every stage.

Sources: V2 RESULTS; V5 RESUMPTION_20261026_RESULTS; V6 RESULTS. Different subsets and protocols; these scores do not form a unified performance curve or establish that the methods fail in general.

Zero on a Subtype Is Not Zero Overall

Appendix: denominators explain apparently conflicting scores

AMBIGREF: V17 baseline

SUPPORTS: **9**/**9** correct
NEI: **0**/**6** correct

Total: $9/15 = 60\%$

All challenge items

Correct: 54 SUPPORTS
+ 19 REFUTES + 28 NEI

$101/141 = 71.63\%$

A subtype's 0% does not imply 0% on the whole mechanism.

Sources: V17 predictions and development variants; Team A reference deck. Team A reports 66.7% overall challenge accuracy on slide 12. Its low-score denominators on slides 8–9 are not explicit. We reproduce related error patterns, not an exact recalculation of those percentages.

Difficult Subtypes: Results by Seed

Appendix: correct counts for the 22-item NEI diagnostic subset

Mechanism / variant	N	Base	NLI	PJ	L42	L43	L44
LOOKALIKE / bare	5	0	0	1	4	5	5
AMBIGREF / diff_gender_other_pronoun	3	0	0	2	3	3	3
UNIQUE / other_indef | indef	2	0	1	1	2	2	2
SETCLOSE / subset_count	2	1	1	2	2	2	1
PRESUP / still_decoy | before	3	0	0	2	3	3	1
TEMPSCOPE / first_term	3	1	1	1	3	3	3
DENIAL / baseless	4	0	1	4	4	4	4
Total	**22**	**2**	**4**	**13**	**21**	**22**	**19**

Base = Baseline. L42 / L43 / L44 = LoRA seeds 42 / 43 / 44.

Source: V17 predictions joined to V16 revision 2 development variants. Post-hoc selected NEI items only; variant fields are not model inputs. Seeds do not create extra independent items. PRESUP includes only the three NEI-labeled cases of the specified variant.

Limitations and Open Questions

Appendix: how to interpret the current results

Evidence limits

- Prior development-data exposure
- AI-assisted rationale review
- No PRESUP training coverage
- Some label interpretations remain disputed

Which method is best?

LoRA has the highest V17 challenge accuracy. PJ alone meets all V17 criteria.

Why did fusion not improve further?

The completed designs did not meet all criteria. The cause has not been isolated.

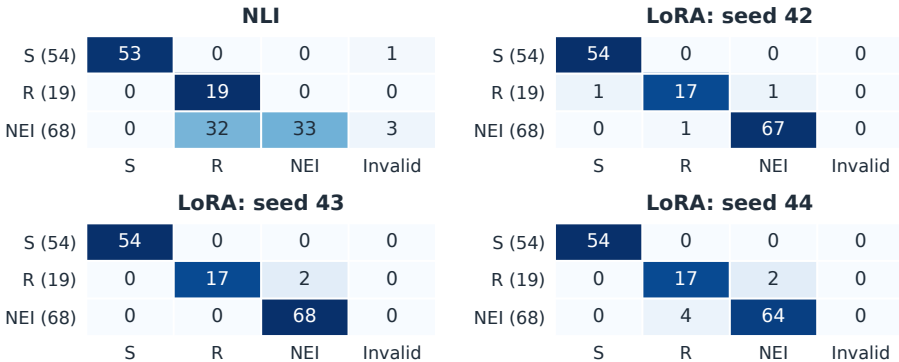
What remains unfinished?

49 full-scope comparison configurations, added post-distillation review, epoch-wise selection and independent evaluation.

Sources: V16 revision 2 DEVIATIONS; V18 quick24 ASSESSMENT. V16 engineering tests read raw full/test files for overlap and deduplication checks. Those reads were not used for this run's model training, selection or evaluation, but strict data-sealing acceptance was not fully satisfied. V18's final five-family selection excluded standalone PJ.

Confusion Matrices by Method and Seed

Appendix: cell text shows item counts; color shows the percentage within each true class.



Rows: true label. Columns: predicted label. S = SUPPORTS; R = REFUTES; NEI = NOT ENOUGH INFO.

Source: V17 frozen predictions. Every panel uses the same 141 challenge items; row counts are 54, 19 and 68. Shared color scale: 0–100%. Invalid outputs count as errors. Each seed is a separate run, not an extra independent set of items.